



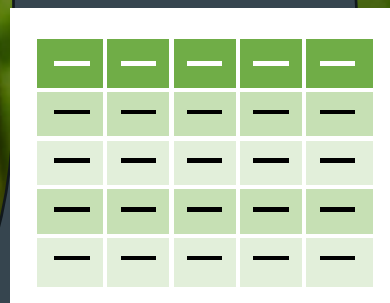


EXCEL

MACROS

+

Introduction to VBA



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VISUAL BASIC FOR APPLICATIONS



- Customize and automate Excel.
- Record macros to reuse sets of instructions without coding.
- Create your own custom Excel functions.
- Create specific routines from scratch using VBA.
- Create input forms.

Warning about using VBA



Workbooks containing VBA code will be saved as .xlsm (macro-enabled) files.



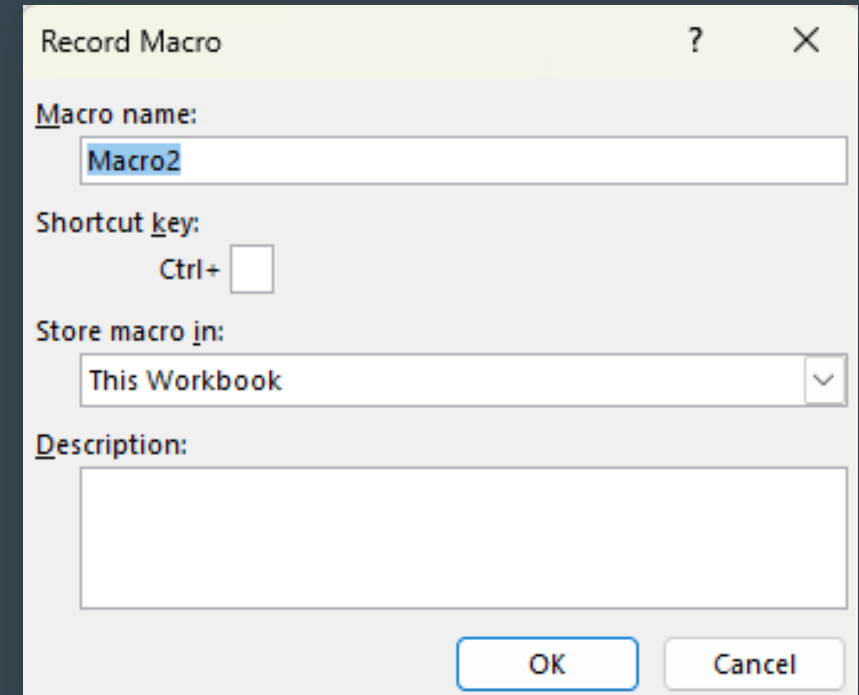
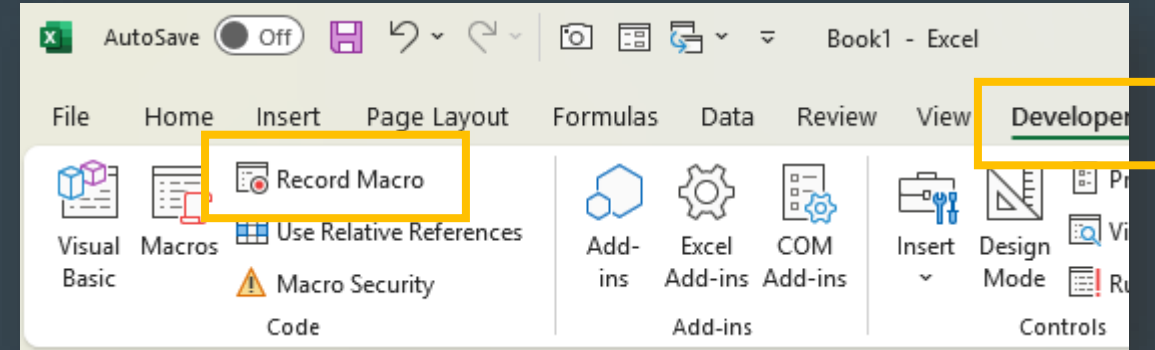
By default Microsoft will prevent .xlsm files to run automatically. You might have to authorize their use (trusted files).



You will need to activate the Developer tab in Excel.
To do so, personnalize any ribbon and activate the developer tab in that window.

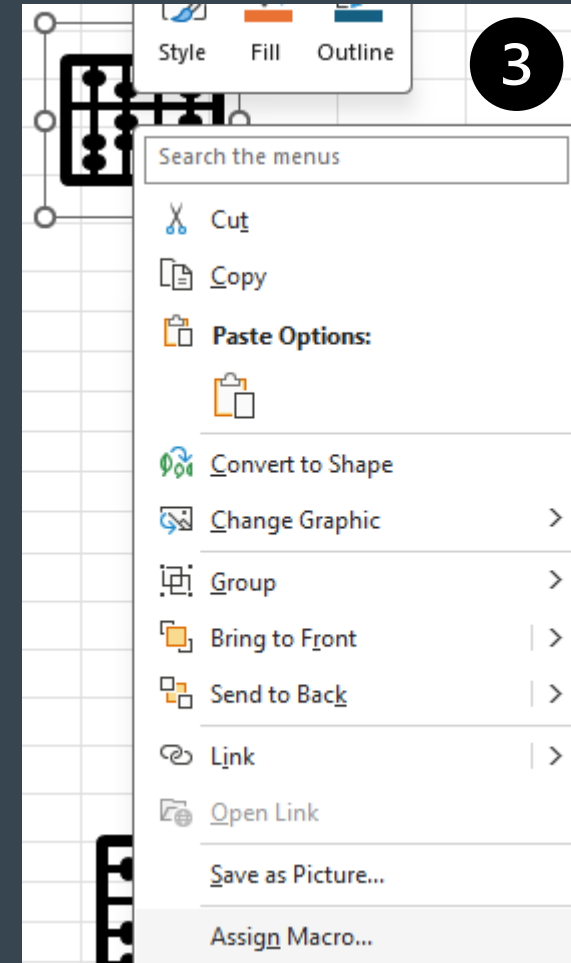
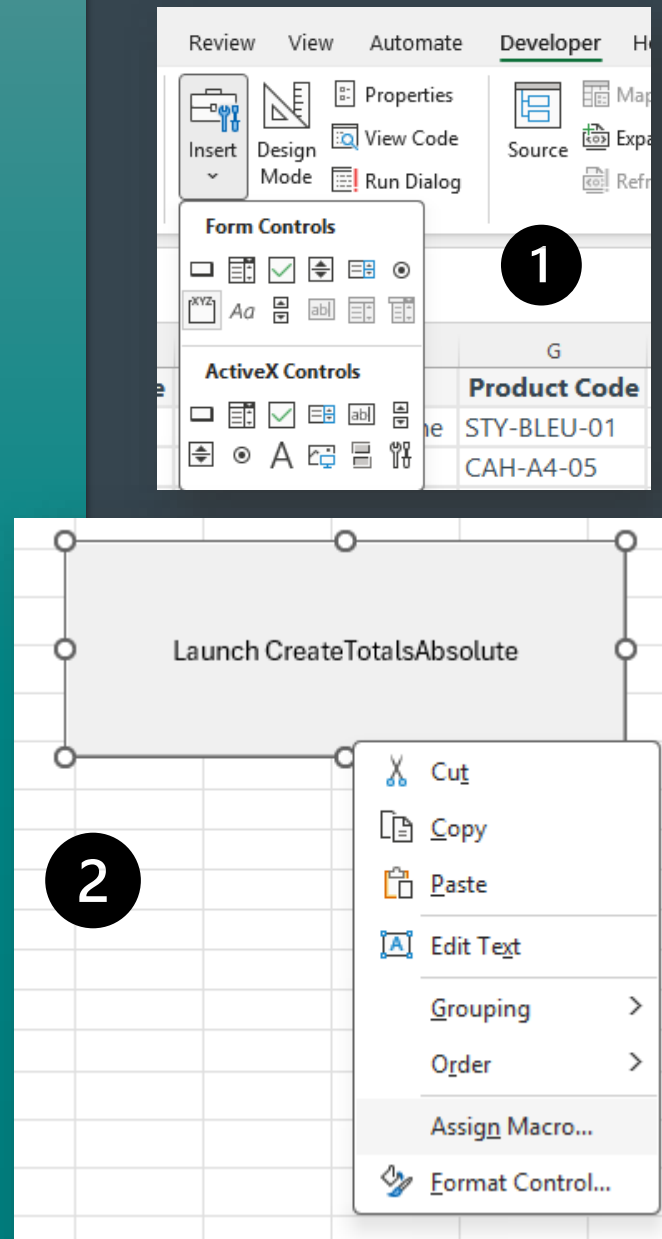
My first macro

- In the developer tab, press 'Record Macro', name it 'CreateTotalsAbsolute' and optionally a shortcut (macro shortcuts override the existing shortcuts).
- Select the cell H2 on the sheet Macro1.
- Type the formula: '=B2*C2' and copy the formula over all rows to H6.
- Select H9 and type this formula: '=SUM(' then select cells H2:H6 and type ')'. Press Enter.
- Press Stop Recording and open the VBE to view the VBA code created by the macro recorder.



Assigning a macro to a button

1. From the developer tab, insert a button in your Excel sheet then right-click it and assign the newly created macro to it.
2. Click the button to launch the macro.
3. Repeat the process for a shape, an icon or a picture.
4. Select H2 and click the button or the icon.
5. Delete the added cells and select H15. Click on the button (or icon) again. What do you see ?



My second macro

- In the developer tab, click on 'Use Relative References' then press 'Record Macro' and name it 'CreateTotalsRelative'.
- Duplicate the exact same steps as in the first macro.
- Press Stop.
- Insert a button and assign this new macro to it.
- Delete all previous calculations and select H2. Click on the button. Now, select H15 and press the same button again. What do you see ?

Product Code	
STY-BLEU-01	60
CAH-A4-05	48
AGD-2024-07	139.3
SRS-USB-03	105
CLV-AZT-04	196
Total	548.3

My third macro

- Go to sheet Macro3-5.
- In the developer tab, click on 'Use Relative References' then press 'Record Macro' and name it 'TotalWithTax' (3rd macro).
- Select I2 then create the formula that calculates row by row the total with tax. The VAT percentage is available in cell B10.
- In I10, calculate the total of I2:I6.
- Press Stop recording.
- Delete all calculations in column I and click on Developer > Macros > TotalWithTax and click 'Run'.

G	H	I
Product Code		
STY-BLEU-01	60	64.86
CAH-A4-05	48	51.888
AGD-2024-07	139.3	150.5833
SRS-USB-03	105	113.505
CLV-AZT-04	196	211.876
Total	548.3	592.7123

My fourth macro

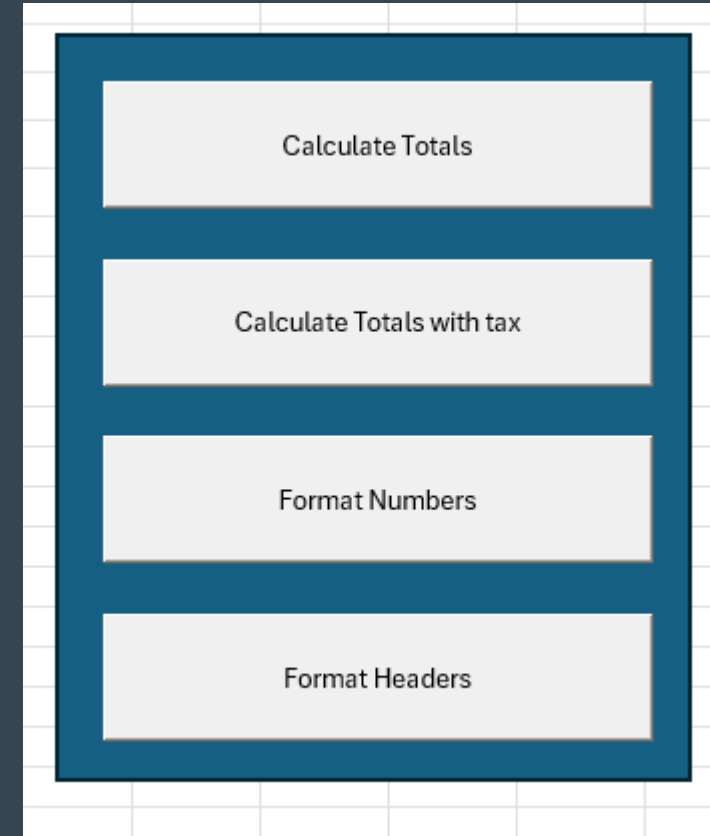
- In sheet Macro3-5.
- Record a 4th macro 'FormatNumbers' that format all cells in Columns H and I in the accounting format with CHF prefix and 0 decimals.
- Stop recording
- Record a 5th macro 'FormatHeaders' that inserts the headers 'Total' and 'TotalwithTax' in H1 & I1 then applies cell formatting to the header rows. It also has to apply bold to all calculations in H & I and sets the font size to 20 for the totals in row 10 (including the header). Stop recording

G	H	I
Product Code		
STY-BLEU-01	60 CHF	65 CHF
CAH-A4-05	48 CHF	52 CHF
AGD-2024-07	139 CHF	151 CHF
SRS-USB-03	105 CHF	114 CHF
CLV-AZT-04	196 CHF	212 CHF
Total	548 CHF	593 CHF

Product	Quantity	Unit Price	Customer Name	Order Date	City	Product Code	TOTAL	TOTAL WITH TAX
Stylo bleu	24	2.5	Martin	45303	Lausanne	STY-BLEU-01	60 CHF	65 CHF
Cahier A4	10	4.8	Dupont	15.01.2024	Genève	CAH-A4-05	48 CHF	52 CHF
Agenda 2024	7	19.9	Pascal	03.02.2024	Fribourg	AGD-2024-07	139 CHF	151 CHF
Souris USB	3	35	Laurent	20.02.2024	Sion	SRS-USB-03	105 CHF	114 CHF
Clavier AZERTY	4	49	Fontaine	28.02.2024	Neuchâtel	CLV-AZT-04	196 CHF	212 CHF
VAT %	8.10%					Total	548 CHF	593 CHF

Summary

- In sheet Macro3–5.
- Insert a rectangle shape and insert 4 buttons into it.
- Assign the 4 buttons to *CreateTotalsRelative*, *TotalWithTax*, *FormatNumbers* and *FormatHeaders*.
- Restore Sheet Macro63–to its original state then apply all 4 macros sequentially to the dataset.



When to use each

✓ Relative references

- Processing data row by row
- Formatting cells around the active cell
- Creating reusable macros

✓ Absolute references

- Updating a dashboard
- Writing to fixed report locations
- Refreshing specific cells/ranges

If unsure, record the same macro twice – once with Developer → Use Relative References enabled and once disabled – and then compare the results with your objectives to see which version best suits your needs.

When Recorded Macros Are Most Valuable

- ✓ Formatting repetitive reports
- ✓ Cleaning imported data
- ✓ Monthly reporting
- ✓ Refreshing dashboards
- ✓ Creating workbook templates
- ✓ Learning VBA by examining the generated code

When Recorded Macros Are NOT the best choice

- ✗ Complex decision-making (If/Then logic)
- ✗ Loops through thousands of rows
- ✗ Dynamic file handling
- ✗ User forms
- ✗ Advanced automation across multiple workbooks

In those cases, hand-written VBA is usually much better.

A good rule: If you can perform the task manually in less than 30 clicks and you repeat it every day or every week, it's an excellent candidate for a recorded macro.

SHARING MACROS ACROSS WORKBOOKS

Option 1: **Store the Macro in the Personal Macro Workbook** (Best for Personal Use)

When recording a macro, choose: Store macro in: Personal Macro Workbook

This saves the macro in a hidden workbook called PERSONAL.XLSB that opens automatically whenever Excel starts.

Benefits: Available in every workbook on your computer. Great for formatting, cleanup, and productivity macros you use regularly.

Limitation: Other users won't have access unless you share the PERSONAL.XLSB file.

Option 2: **Create a Macro Library Workbook**

Save commonly used macros in a dedicated workbook such as 'CompanyMacros.xlsm'.

Open it alongside your working files.

Benefits:

- Easy to maintain a central collection of macros.
- Multiple users can work from the same macro library.

COPYING MACROS BETWEEN WORKBOOKS

- Press Alt + F11 to open the VBA Editor.
- Expand the source workbook.
- Open Modules.
- Drag the module to the target workbook, or export/import it.
- You can also copy/paste the VBA code inside modules between workbooks.



- Microsoft Quick Start: Create a macro
<https://support.microsoft.com/en-us/excel/get-started-with-excel/quick-start-create-a-macro>

Your feedback

<https://learn.digicomp.ch>

<https://digicomp.my.salesforce-sites.com/EKBF?lang=fr>

